Expenzo: A Verifiable Web Architecture for Personal Finance Intelligence, Deterministic Accounting, and Multi-Tenant Security

Full-Stack Implementation, Atomic Financial RPCs, and Empirical Test Evaluation

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Status: Ready for Peer-Review Submission (Formal Review Pending Faculty Advisor Sign-Off)

Abstract

Personal financial tracking tools frequently suffer from computational drift caused by binary floating-point representation, severe data leakage risks in multi-tenant environments lacking database-level isolation, and hallucination-prone advice from open-ended language models. This paper presents **Expenzo**, a verified web application engineered to solve these challenges. Expenzo employs a strict 64-bit integer paise accounting model that eliminates rounding errors at the database kernel. The system implements a three-tier architecture utilizing React 19, TypeScript, and a Supabase PostgreSQL backend protected by Row Level Security (RLS) and atomic stored procedures (execute_transfer, pay_bill). A hybrid co-pilot pairs a sub-50ms deterministic 11-intent regex parser with a secure server-side Gemini Flash Edge Function. The implementation achieved a 100% pass rate across 184 tests in 20 suites covering data validation, atomic state transitions, and user isolation.

Keywords: Personal Finance Management, Deterministic Bookkeeping, Row Level Security, Integer Arithmetic, Full-Stack Architecture, Atomic Stored Procedures.

1. Introduction & Behavioral Foundations

Behavioral economics shows that consumers organize expenditures into distinct mental accounts (Thaler, 1985). Uncoupling payment mechanisms reduces spending salience (Soman, 2001). Expenzo addresses this via real-time budget envelopes, zero-drift 64-bit integer paise calculations, and kernel-enforced Row Level Security in PostgreSQL.

2. System Architecture & Relational Schema

The system is structured into three tiers: (1) Presentation Layer with React 19 and Zustand; (2) Logic & Edge Layer with deterministic regex intent parsing (<50ms) and Supabase Edge Functions; (3) Persistence Layer with Supabase PostgreSQL 15 hosting 17 RLS-protected tables.

Entity	Primary Key	Foreign Key	Integrity Rule
profiles	id (UUID)	auth.users(id)	User identity, currency, persona
financial_accounts	id (UUID)	profiles(id)	Type check, opening balance minor
categories	id (UUID)	profiles(id)	Unique(user_id, name), budget minor
transactions	id (UUID)	accounts(id)	Canonical ledger, paise integer arithmetic
recurring_bills	id (UUID)	categories(id)	Due date recurrence, auto-settle RPC

3. Empirical Verification & Automated Test Battery

The platform was verified via 184 automated tests across 20 test suites with a 100% pass rate. Testing covered store immutability, integer math precision, Row Level Security boundary enforcement, atomic transfer execution with rollback, and deterministic AI parsing latency (averaging 18ms).

4. Conclusion & Submission Sign-Off

Expenzo proves that personal financial management can combine mathematical exactness, database-level security isolation, and privacy-preserving AI assistance. The complete platform is live and verified.

Academic Status: Manuscript fully synchronized with code implementation. Ready for formal journal/conference submission pending faculty advisor sign-off.